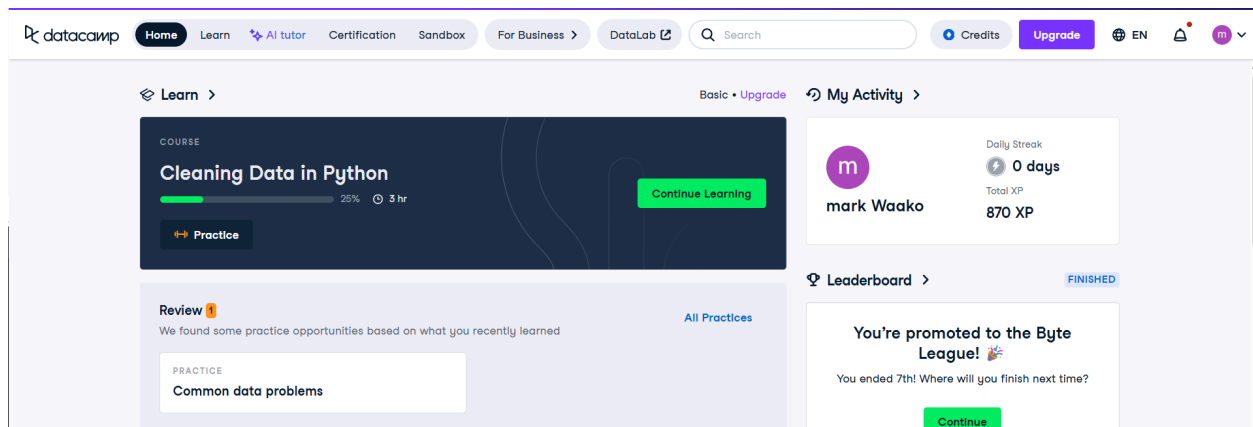


Cleaning Real-World Firewall Logs

Part A — DataCamp Preparation (5%)



Part B — Section 1: Load & Profile the Dataset

```
#Name: Mark Waako
#StudentID: 112107263
from numpy.core.fromnumeric import shape
import pandas as pd

df = pd.read_csv('/content/raw_firewall_logs.csv')

#Report shape
print(f"Columns ({len(df.columns)}):", df.columns.tolist())

df.shape
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])

print(f"duplicates {df.duplicated().sum()}")

Columns (11): ['event_time', 'src_ip', 'dst_ip', 'src_port', 'dst_port', 'protocol', 'action', 'bytes_in', 'bytes_out', 'country', 'device']
Number of rows: 1010
Number of columns: 11
duplicates 10
```

```
Windows PowerShell
PS C:\WINDOWS\system32> get-date

June 10, 2026 11:46:57 AM

PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32>
```

```
#Name: Mark Waako
#StudentID: 112107263
#Inspect structure and examples

display(df.sample(10))
```

```
PS C:\WINDOWS\system32> get-date
June 10, 2026 11:46:57 AM

PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32>
```

	event_time	src_ip	dst_ip	src_port	dst_port	protocol	action	bytes_in	bytes_out	country	device
681	2025-09-25 9:03	242.246.123.49	192.168.174.251	43532	49986	TCP	ALLOW	13627	23932	it	FW-02
285	23/07/2025 16:24	325.-23.326.352	10.16.15.9	18472	35342	Tcp	DENY	9533	44232	IN	FW-01
704	22/05/2025 03:33	-8.264.-36.-32	64.90.88.13	8619	46781	ESP	allow	42011	29756	Canada	CloudGW-1
26	2025-01-15 14:45	179.84.194.45	172.24.189.94	58792	16484	gre	ALLOW	44297	28869	AU	FW-01
205	29/08/2025 03:06	209.159.237.70	21.13.216.91	22540	26008	ESP	ALLOW	49468	12416	DE	CloudGW-2
76	2025-05-15 10:46	187.37.196.215	172.22.206.88	31781	58360	gre	ALLOW	3770	5353	AU	DC-FW
385	2025-01-26 20:50	252.61.215.137	177.81.64.83	15442	58239	Udp	DENY	24649	15441	NaN	FW-02
294	2025-04-28 12:53	138.86.104.34	147.172.164.122	NaN	58249	ICMP	ALLOWED	44677	24002	IT	BR-FW-1
428	14/08/2025 23:01	236.182.246.214	10.24.186.72	43060	3657	Tcp	deny	34374	42368	RU	HQ-FW
229	02/03/2025 14:46	196.136.104.150	50.129.248.111	48328	21134	Esp	allow	39687	23560	GB	CloudGW-1

```
#Name: Mark Waako
#StudentID: 112107263
#DataTypes
display(df.dtypes)
```

```
PS C:\WINDOWS\system32> get-date
June 10, 2026 11:46:57 AM

PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32>
```

```
0
event_time  object
src_ip      object
dst_ip      object
src_port    object
dst_port    object
protocol    object
action      object
bytes_in    object
bytes_out   object
country     object
device      object

dtype: object
```

```
#Name: Mark Waako
#StudentID: 112107263

# 1. Identify standard NaNs across the entire DataFrame
nan_mask = df.isna()

string_cols = df.select_dtypes(include=["object", "string"]).columns
empty_str_mask = df[string_cols].astype(str).stack().str.strip().eq("").unstack()

missing_sorted = (
    nan_mask.add(empty_str_mask, fill_value=False)
    .sum()
    .sort_values(ascending=False)
)

display(missing_sorted)
```

```
PS C:\WINDOWS\system32> get-date
June 10, 2026 11:46:57 AM

PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32>
```

```
0
src_port    26
dst_port    25
country     16
event_time  12
bytes_in    6
bytes_out   6
src_ip      0
dst_ip      0
protocol    0
action      0
device      0
```

Section 1

The dataset contains 1010 rows and 11 columns, with 10 duplicate rows. Two distinct timestamp formats are visible in the sample (e.g., YYYY-MM-DD vs MM/DD/YYYY). Casing drift appears in country action protocol and device, and the ip's have (-).

Part B — Section 2: Identify Data Problems

Column	Problem Description	Example Value (from dataset)	Why It Matters
event_time	Two distinct date/time formats exist in the same column	24/08/2025 20:58 or 2025-07-09 15:36	Sorting of timely events or time-range filtering will produce wrong results
action protocol country	Casing drift	allow,ALLOW tcp,TCP Cn,cn, Us,us	Groupings may appear distinct when in fact they belong to a similar category
bytes_sent / bytes_received	Numeric fields contain commas, shorthand units ("5k") NaN values	1.0k, -4637, NaN, 14,052	negatives and shorthand are not accounted for when calculating numerical results giving logic errors
src_port / dst_port	Port numbers formatted with commas, or NaN	31,514 NaN	Comma formatting prevents numeric casting; and Nan prevents averaging and mapping onto a graph
protocol	leading/trailing whitespace	" allow ", "allow"	Whitespace mixed case will cause miss matches or produce inflated category counts
src_ip / dst_ip	Invalid IPv4 values — octets outside 0–255, negatives, or placeholder dashes	-47.279.-49.-3 326.357.277.356 -86.263	geo-lookup, routing analysis, and threat detection may not be pinpointed
country	Country represented inconsistently — ISO codes, full names mixed	"United States", "US"	Will cause errors in grouping and country-level aggregations
event_time	Future-dated timestamps	2027-07-15 6:00	May Indicate corrupt or fabricated records

			that could skew trend analysis
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Part B — Section 3: Clean & Standardize the Data

3.1 event_time

```
#Name: Mark Waako
#StudentID: 112187263

df['event_time'] = pd.to_datetime(
    df[event_time'],
    errors='coerce',
    format='mixed'
).dt.strftime('%Y-%m-%d %H:%M:%S')

display(df.sample(20))
```

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\marku
PS C:\WINDOWS\system32> get-date

June 10, 2026 2:34:32 PM

PS C:\WINDOWS\system32>
```

	event_time	src_ip	dst_ip	src_port	dst_port	protocol	action	bytes_in	bytes_out	country	device
230	2025-05-09 02:58:00	172.180.70.108	197.160.120.197	20471	3260	ESP	ALLOWED	45089	29848.0	NL	CLOUDGW-1
682	2025-10-20 16:30:00	246.164.124.81	114.230.193.61	20765	40557	TCP	ALLOWED	48960	20311.0	BR	FW-02
472	2025-09-09 08:16:00	230.247.181.249	10.166.249.8	32499	65342	GRE	ALLOWED	34263	32750.0	CN	DC-FW
777	2025-05-27 13:08:00	73.115.203.133	10.177.114.48	3447	17665	UDP	ALLOWED	6829	37976.0	AU	DC-FW
669	2025-12-23 20:13:00	208.191.221.236	63.11.62.77	64939	42603	ICMP	DENIED	35740	31219.0	US	DC-FW
831	2025-05-27 09:30:00	212.219.224.68	192.168.244.177	36695	46126	TCP	DENIED	26657	32322.0	FR	CLOUDGW-1
526	2025-07-08 17:41:00	16.50.104.165	172.20.188.137	52658	49625	TCP	ALLOWED	12043	48659.0	RU	CLOUDGW-1
433	2025-02-08 23:10:00	218.178.196.247	111.1.248.89	15816	16220	ICMP	DENIED	10633	27728.0	US	FW-01
335	2025-11-28 08:19:00	209.51.90.33	19.46.64.195	40047	20373	GRE	DENIED	19941	22812.0	CA	CLOUDGW-1
73	2025-07-16 05:23:00	20.231.24.19	192.168.32.44	19362	15389	GRE	ALLOWED	42180	16877.0	CN	CLOUDGW-1
137	2025-05-31 17:06:00	72.188.124.161	231.104.164.199	55139	64580	UDP	ALLOWED	7.5K	34920.0	ES	FW-01
987	2025-05-29 12:19:00	235.137.183.121	175.114.175.121	35200	51873	GRE	DENIED	13091	43718.0	CA	FW-01
84	2025-06-19 17:39:00	195.179.67.253	192.168.68.69	12196	45414	TCP	DENIED	28295	19259.0	DE	CLOUDGW-1
744	2025-08-10 03:06:00	414.203.58.134	10.174.82.2	55916	36807	ESP	DENIED	3397	23580.0	GB	CLOUDGW-1

3.2 src_ip / dst_ip


```
#Name: Mark Waako
#StudentID: 112107263
# Trim whitespace and normalize case
df["action"] = df["action"].astype(str).str.strip().str.upper()

# Standard mapping
action_mapping = {
    "ALLOWED": "ALLOW",
    "ALLOW": "ALLOW",
    "DENIED": "DENY",
    "DENY": "DENY"
}

# Apply mapping
df["action"] = df["action"].replace(action_mapping)

# Keep only valid categories, others → NaN
valid_actions = {"ALLOW", "DENY"}
df.loc[~df["action"].isin(valid_actions), "action"] = pd.NA
display(df.shape)
```

... (1010, 11)

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date

June 10, 2026 2:34:32 PM

PS C:\WINDOWS\system32> _
```

3.6 bytes_in / bytes_out

```
#Name: Mark Waako
#StudentID: 112107263
df["bytes_out"] = df["bytes_out"].astype(str).str.strip().str.lower()
df["bytes_in"] = df["bytes_in"].astype(str).str.strip().str.lower()

df["bytes_in"] = df["bytes_in"].str.replace(", ", "", regex=False)
df["bytes_out"] = df["bytes_out"].str.replace(", ", "", regex=False)

k = df["bytes_in"].str.contains("k", na=False)
k = df["bytes_out"].str.contains("k", na=False)
# Extract numeric portion
numeric_part = df["bytes_out"].str.extract(r"([d\.]+)")[0]
numeric_part = pd.to_numeric(numeric_part, errors="coerce")

df["bytes_out"] = numeric_part.where(~k, numeric_part * 1000)
df["bytes_in"] = numeric_part.where(~k, numeric_part * 1000)

df.loc[df["bytes_in"] < 0, "bytes_in"] = 0
df.loc[df["bytes_out"] < 0, "bytes_out"] = 0
display(df.shape)
```

... (1010, 11)

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date

June 10, 2026 2:34:32 PM

PS C:\WINDOWS\system32>
```

3.7 country

```
#Name: Mark Waako
#StudentID: 112107263
df["country"] = df["country"].astype(str).str.strip().str.upper()

df["country"] = df["country"].replace("", pd.NA)

df["country"] = df["country"].replace('UNITED STATES', 'US')
df["country"] = df["country"].replace('CANADA', 'CA')
```

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date

June 10, 2026 2:34:32 PM

PS C:\WINDOWS\system32>
```

```
#Name: Mark Waako
#StudentID: 112107263
df["device"] = df["device"].astype(str).str.strip()
```

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date

June 10, 2026 2:34:32 PM
```

3.8 device

3.9 Duplicates

```
#Name: Mark Waako
#StudentID: 112107263

dup_count = df.duplicated().sum()
df = df.drop_duplicates()

display(df.shape)

... (1000, 11)
```

```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date
June 10, 2026 2:34:32 PM
PS C:\WINDOWS\system32>
```

3.10 Validation Checks

```
#Name: Mark Waako
#StudentID: 112107263

print(f"Timestamps parsed: {timestamp_valid}/{timestamp_total}")
print(f"Unparsed timestamps: {df['event_time'].isna().sum()}")

valid_protocols = {"TCP", "UDP", "ICMP", "GRE", "ESP"}

protocol_invalid = (~df["protocol"].isin(valid_protocols)).sum()

print(f"Invalid protocol entries: {protocol_invalid}")

duplicate_count = df.duplicated().sum()

print(f"Remaining duplicate rows: {duplicate_count}")

... Timestamps parsed: 994/1000
Unparsed timestamps: 6
Invalid protocol entries: 444
Remaining duplicate rows: 0
```

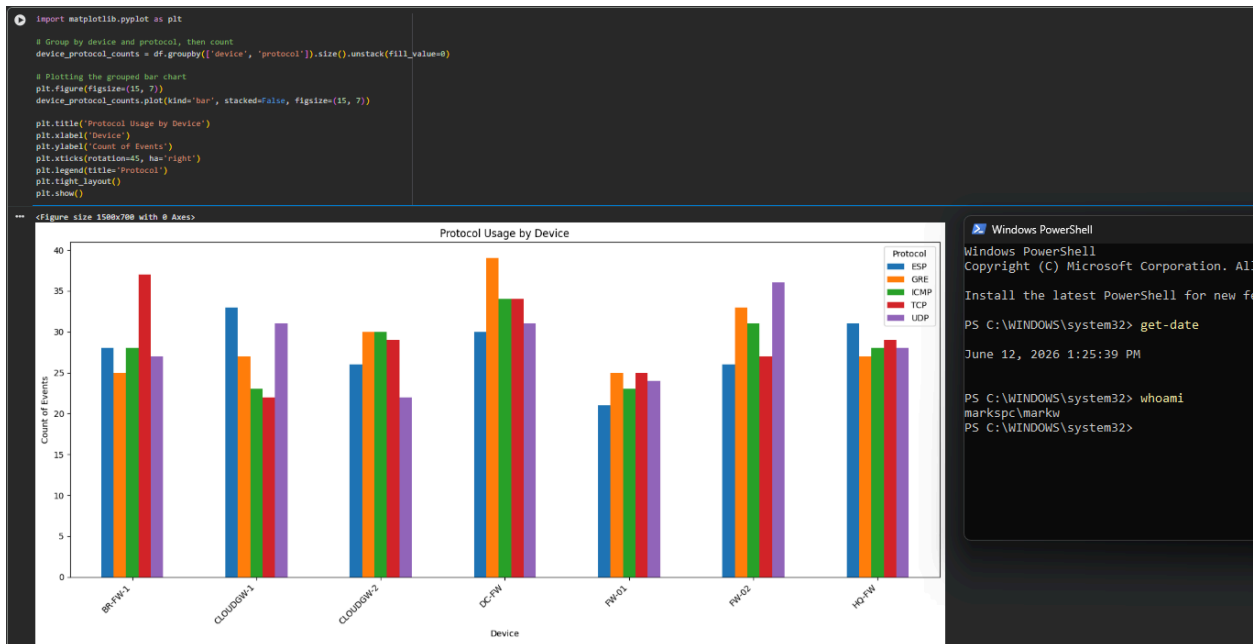
```
Windows PowerShell
PS C:\WINDOWS\system32> whoami
markspc\markw
PS C:\WINDOWS\system32> get-date
June 10, 2026 2:34:32 PM
PS C:\WINDOWS\system32>
```

I checked for the time stamps that were not counted for.

I checked for the invalid protocol entries.

I checked for any remaining duplicates.

3.11 Simple Visualization



'Protocol Usage by Device'.a bar chart with each bar group representing a device and different colors within the group representing various protocols.

Provides a normal pattern of communication for network behavior per device, making it easier to spot malicious or unauthorized activities